

RESEARCH ONLY — NOT ACCEPTED FOR DESIGN

Predictive N_T Replay Report

Project 223 | Design 256 | Source job ID: a72e53d4-6747-4bc1-a36f-80bffd2dc18

Executive conclusion

All 10 N_T systems numerically closed and reproduced their recorded branch. Replay trials N_T=1 and N_T=2 have no replay-stage post-split TPD blocker; they are not accepted design results. N_T=3 through N_T=10 fail post-split TPD stability and are rejected fail-closed.

Accepted Predictive N_T: NOT ASSIGNED. Formal COSMO-SAC sulfur: NOT CALCULABLE. No release or design decision may be inferred from this replay.

Configuration and stage convention

Stage numbering is bottom-to-top: Stage 1 = RRBO-feed/bottom; final stage = fresh-NMP/top. Therefore Stage 1 is the bottom RRBO-feed endpoint; for each N_T, Stage N_T is the top fresh-NMP endpoint.

Fresh Stage 1 flash (Task206 vector)

Metric	Value
Two-phase flash acceptance	ACCEPTED numerical flash
Optimizer / equilibrium polish	success / success
Polish residual / extract fraction	2.8422e-14 / 0.440490
Pre-split minimum TPD / verdict	-0.207414 / UNSTABLE_NEGATIVE_TPD
Post-split TPD minima (R / E)	1.6293e-17 / 5.8970e-18
Max composition separation / material balance residual	0.771341 / 1.1102e-16

N_T=1–10 replay comparison

N_T	Max scaled resid	Worst stage/phase TPD	Stability verdict	NMP-free recovery	Saturates	Total aromatics	Polar aromatics	NMP in raffinate	Targets
1	2.53e-14	S1/R -1.65e-17	No replay-stage TPD blocker	91.30%	89.98%	9.03%	0.99%	7.27%	1 pass / 4 fail / 1 NC
2	5.22e-14	S1/E -3.41e-18	No replay-stage TPD blocker	88.79%	92.02%	7.40%	0.58%	6.98%	2 pass / 3 fail / 1 NC
3	7.68e-14	S1/R -2.43e-2	FAIL: post-split TPD	87.50%	93.10%	6.54%	0.36%	6.83%	3 pass / 2 fail / 1 NC
4	3.71e-14	S1/E -3.69e-2	FAIL: post-split TPD	86.75%	93.75%	6.02%	0.24%	6.75%	3 pass / 2 fail / 1 NC
5	2.89e-14	S1/E -4.39e-2	FAIL: post-split TPD	86.27%	94.16%	5.68%	0.16%	6.69%	3 pass / 2 fail / 1 NC
6	8.42e-14	S1/R -4.82e-2	FAIL: post-split TPD	85.95%	94.44%	5.45%	0.11%	6.66%	3 pass / 2 fail / 1 NC
7	1.22e-13	S1/R -5.11e-2	FAIL: post-split TPD	85.72%	94.63%	5.29%	0.08%	6.63%	3 pass / 2 fail / 1 NC
8	3.20e-13	S1/E -5.31e-2	FAIL: post-split TPD	85.56%	94.77%	5.17%	0.05%	6.61%	3 pass / 2 fail / 1 NC
9	5.33e-12	S1/R -5.46e-2	FAIL: post-split TPD	85.45%	94.87%	5.09%	0.04%	6.60%	3 pass / 2 fail / 1 NC

N_T	Max scaled resid	Worst stage/ phase TPD	Stability verdict	NMP-free recovery	Saturates	Total aromatics	Polar aromatics	NMP in raffinate	Targets
10	8.36e-12	S1/E -5.57e-2	FAIL: post-split TPD	85.36%	94.95%	5.03%	0.03%	6.59%	3 pass / 2 fail / 1 NC

Task206 nine-parameter vector

Parameter	Value
A_MONO_NMP_asymmetry	-0.550455907497
A_MONO_NMP_ref	2.308305427722
A_MONO_NMP_temperature	-0.680381006538
A_SAT_MONO_asymmetry	-0.673709197592
A_SAT_MONO_ref	3.431076990665
A_SAT_MONO_temperature	-0.588130214663
A_SAT_NMP_asymmetry	-0.573242605597
A_SAT_NMP_ref	2.913902338473
A_SAT_NMP_temperature	-0.226748917363

Target-compliance detail by N_T

N_T	Target	Calculated	Limit	Status
1	maximum Nmp Raffinate Wt	7.2715	6.5000	FAIL
1	maximum Raffinate Polar Aromatics Wt	0.9911	0.5000	FAIL
1	maximum Raffinate Total Aromatics Wt	9.0292	5.0000	FAIL
1	minimum Nmp Free Recovery Pct	91.2987	85.0000	PASS
1	minimum Raffinate Saturates Wt	89.9797	90.0000	FAIL
1	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
2	maximum Nmp Raffinate Wt	6.9811	6.5000	FAIL
2	maximum Raffinate Polar Aromatics Wt	0.5778	0.5000	FAIL
2	maximum Raffinate Total Aromatics Wt	7.3995	5.0000	FAIL
2	minimum Nmp Free Recovery Pct	88.7898	85.0000	PASS
2	minimum Raffinate Saturates Wt	92.0227	90.0000	PASS
2	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
3	maximum Nmp Raffinate Wt	6.8347	6.5000	FAIL
3	maximum Raffinate Polar Aromatics Wt	0.3633	0.5000	PASS
3	maximum Raffinate Total Aromatics Wt	6.5354	5.0000	FAIL
3	minimum Nmp Free Recovery Pct	87.5042	85.0000	PASS
3	minimum Raffinate Saturates Wt	93.1013	90.0000	PASS
3	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
4	maximum Nmp Raffinate Wt	6.7491	6.5000	FAIL
4	maximum Raffinate Polar Aromatics Wt	0.2384	0.5000	PASS
4	maximum Raffinate Total Aromatics Wt	6.0159	5.0000	FAIL
4	minimum Nmp Free Recovery Pct	86.7484	85.0000	PASS
4	minimum Raffinate Saturates Wt	93.7458	90.0000	PASS
4	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
5	maximum Nmp Raffinate Wt	6.6948	6.5000	FAIL

N_T	Target	Calculated	Limit	Status
5	maximum Raffinate Polar Aromatics Wt	0.1606	0.5000	PASS
5	maximum Raffinate Total Aromatics Wt	5.6794	5.0000	FAIL
5	minimum Nmp Free Recovery Pct	86.2673	85.0000	PASS
5	minimum Raffinate Saturates Wt	94.1600	90.0000	PASS
5	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
6	maximum Nmp Raffinate Wt	6.6583	6.5000	FAIL
6	maximum Raffinate Polar Aromatics Wt	0.1101	0.5000	PASS
6	maximum Raffinate Total Aromatics Wt	5.4508	5.0000	FAIL
6	minimum Nmp Free Recovery Pct	85.9450	85.0000	PASS
6	minimum Raffinate Saturates Wt	94.4391	90.0000	PASS
6	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
7	maximum Nmp Raffinate Wt	6.6329	6.5000	FAIL
7	maximum Raffinate Polar Aromatics Wt	0.0764	0.5000	PASS
7	maximum Raffinate Total Aromatics Wt	5.2901	5.0000	FAIL
7	minimum Nmp Free Recovery Pct	85.7214	85.0000	PASS
7	minimum Raffinate Saturates Wt	94.6335	90.0000	PASS
7	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
8	maximum Nmp Raffinate Wt	6.6148	6.5000	FAIL
8	maximum Raffinate Polar Aromatics Wt	0.0535	0.5000	PASS
8	maximum Raffinate Total Aromatics Wt	5.1744	5.0000	FAIL
8	minimum Nmp Free Recovery Pct	85.5622	85.0000	PASS
8	minimum Raffinate Saturates Wt	94.7722	90.0000	PASS
8	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
9	maximum Nmp Raffinate Wt	6.6015	6.5000	FAIL
9	maximum Raffinate Polar Aromatics Wt	0.0377	0.5000	PASS
9	maximum Raffinate Total Aromatics Wt	5.0893	5.0000	FAIL
9	minimum Nmp Free Recovery Pct	85.4465	85.0000	PASS
9	minimum Raffinate Saturates Wt	94.8730	90.0000	PASS
9	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE
10	maximum Nmp Raffinate Wt	6.5916	6.5000	FAIL
10	maximum Raffinate Polar Aromatics Wt	0.0267	0.5000	PASS
10	maximum Raffinate Total Aromatics Wt	5.0259	5.0000	FAIL
10	minimum Nmp Free Recovery Pct	85.3612	85.0000	PASS
10	minimum Raffinate Saturates Wt	94.9474	90.0000	PASS
10	target Raffinate Sulfur Ppm	NOT CALCULABLE	1000.0000	NOT_CALCULABLE

Appendix A — stage-level post-split stability

Every row reports the recorded post-split TPD-search minimum and local chemical-potential-Jacobian minimum eigenvalue for one phase. Negative TPD is an explicit fail-closed blocker.

N_T	Stage	Phase	TPD minimum	TPD verdict	Local min. eigenvalue	Local verdict
1	1	Raffinate	-1.65447e-17	STABLE_NO_NEGATIVE_TPD	3.48631e-3	—
1	1	Extract	4.25420e-18	STABLE_NO_NEGATIVE_TPD	5.36014e-3	—
2	1	Raffinate	1.32882e-18	STABLE_NO_NEGATIVE_TPD	4.50508e-3	—

N_T	Stage	Phase	TPD minimum	TPD verdict	Local min. eigenvalue	Local verdict
2	1	Extract	-3.41239e-18	STABLE_NO_NEGATIVE_TPD	6.67818e-3	—
2	2	Raffinate	1.67355e-18	STABLE_NO_NEGATIVE_TPD	2.07686e-3	—
2	2	Extract	7.64336e-18	STABLE_NO_NEGATIVE_TPD	3.22657e-3	—
3	1	Raffinate	-2.42752e-2	UNSTABLE_NEGATIVE_TPD	4.96739e-3	—
3	1	Extract	-2.42752e-2	UNSTABLE_NEGATIVE_TPD	7.10740e-3	—
3	2	Raffinate	-1.11484e-17	STABLE_NO_NEGATIVE_TPD	2.93690e-3	—
3	2	Extract	5.82207e-18	STABLE_NO_NEGATIVE_TPD	4.50353e-3	—
3	3	Raffinate	2.38435e-18	STABLE_NO_NEGATIVE_TPD	1.32028e-3	—
3	3	Extract	1.04546e-17	STABLE_NO_NEGATIVE_TPD	2.05375e-3	—
4	1	Raffinate	-3.69368e-2	UNSTABLE_NEGATIVE_TPD	5.21650e-3	—
4	1	Extract	-3.69368e-2	UNSTABLE_NEGATIVE_TPD	7.25292e-3	—
4	2	Raffinate	1.03974e-17	STABLE_NO_NEGATIVE_TPD	3.38517e-3	—
4	2	Extract	5.86376e-18	STABLE_NO_NEGATIVE_TPD	5.12970e-3	—
4	3	Raffinate	9.03568e-17	STABLE_NO_NEGATIVE_TPD	1.96714e-3	—
4	3	Extract	-2.20335e-18	STABLE_NO_NEGATIVE_TPD	3.03306e-3	—
4	4	Raffinate	1.28124e-17	STABLE_NO_NEGATIVE_TPD	8.71556e-4	—
4	4	Extract	-7.88676e-18	STABLE_NO_NEGATIVE_TPD	1.35624e-3	—
5	1	Raffinate	-4.38881e-2	UNSTABLE_NEGATIVE_TPD	5.36332e-3	—
5	1	Extract	-4.38881e-2	UNSTABLE_NEGATIVE_TPD	7.29797e-3	—
5	2	Raffinate	-1.95240e-2	UNSTABLE_NEGATIVE_TPD	3.64444e-3	—
5	2	Extract	-1.95240e-2	UNSTABLE_NEGATIVE_TPD	5.46582e-3	—
5	3	Raffinate	1.90262e-18	STABLE_NO_NEGATIVE_TPD	2.33406e-3	—
5	3	Extract	-4.99336e-18	STABLE_NO_NEGATIVE_TPD	3.57528e-3	—
5	4	Raffinate	8.23922e-18	STABLE_NO_NEGATIVE_TPD	1.34248e-3	—
5	4	Extract	3.25462e-18	STABLE_NO_NEGATIVE_TPD	2.07254e-3	—
5	5	Raffinate	-1.70854e-17	STABLE_NO_NEGATIVE_TPD	5.89456e-4	—
5	5	Extract	4.41353e-18	STABLE_NO_NEGATIVE_TPD	9.17283e-4	—
6	1	Raffinate	-4.81886e-2	UNSTABLE_NEGATIVE_TPD	5.45441e-3	—
6	1	Extract	-4.81886e-2	UNSTABLE_NEGATIVE_TPD	7.30566e-3	—
6	2	Raffinate	-3.09519e-2	UNSTABLE_NEGATIVE_TPD	3.80343e-3	—
6	2	Extract	-3.09519e-2	UNSTABLE_NEGATIVE_TPD	5.65642e-3	—
6	3	Raffinate	-6.08823e-3	UNSTABLE_NEGATIVE_TPD	2.55695e-3	—
6	3	Extract	-6.08823e-3	UNSTABLE_NEGATIVE_TPD	3.89644e-3	—
6	4	Raffinate	-3.59525e-18	STABLE_NO_NEGATIVE_TPD	1.62456e-3	—
6	4	Extract	-4.10782e-19	STABLE_NO_NEGATIVE_TPD	2.49587e-3	—
6	5	Raffinate	6.48987e-18	STABLE_NO_NEGATIVE_TPD	9.28288e-4	—
6	5	Extract	-2.20040e-18	STABLE_NO_NEGATIVE_TPD	1.43362e-3	—
6	6	Raffinate	2.49913e-18	STABLE_NO_NEGATIVE_TPD	4.04779e-4	—
6	6	Extract	1.97722e-18	STABLE_NO_NEGATIVE_TPD	6.30515e-4	—
7	1	Raffinate	-5.10682e-2	UNSTABLE_NEGATIVE_TPD	5.51344e-3	—
7	1	Extract	-5.10682e-2	UNSTABLE_NEGATIVE_TPD	7.29926e-3	—
7	2	Raffinate	-3.79788e-2	UNSTABLE_NEGATIVE_TPD	3.90575e-3	—
7	2	Extract	-3.79788e-2	UNSTABLE_NEGATIVE_TPD	5.76860e-3	—

N_T	Stage (bottom/top)	Phase	TPD minimum	TPD verdict	Local min. eigenvalue	Local verdict
7	3	Raffinate	-2.09928e-2	UNSTABLE_NEGATIVE_TPD	2.69832e-3	—
7	3	Extract	-2.09928e-2	UNSTABLE_NEGATIVE_TPD	4.09488e-3	—
7	4	Raffinate	2.03014e-16	STABLE_NO_NEGATIVE_TPD	1.80206e-3	—
7	4	Extract	1.57869e-18	STABLE_NO_NEGATIVE_TPD	2.75874e-3	—
7	5	Raffinate	-1.72543e-17	STABLE_NO_NEGATIVE_TPD	1.13842e-3	—
7	5	Extract	1.06668e-17	STABLE_NO_NEGATIVE_TPD	1.75157e-3	—
7	6	Raffinate	4.84050e-18	STABLE_NO_NEGATIVE_TPD	6.47658e-4	—
7	6	Extract	4.84464e-18	STABLE_NO_NEGATIVE_TPD	1.00063e-3	—
7	7	Raffinate	6.63419e-18	STABLE_NO_NEGATIVE_TPD	2.81671e-4	—
7	7	Extract	-6.58206e-18	STABLE_NO_NEGATIVE_TPD	4.38221e-4	—
8	1	Raffinate	-5.31058e-2	UNSTABLE_NEGATIVE_TPD	5.55208e-3	—
8	1	Extract	-5.31058e-2	UNSTABLE_NEGATIVE_TPD	7.28853e-3	—
8	2	Raffinate	-4.26598e-2	UNSTABLE_NEGATIVE_TPD	3.97249e-3	—
8	2	Extract	-4.26598e-2	UNSTABLE_NEGATIVE_TPD	5.83599e-3	—
8	3	Raffinate	-3.01681e-2	UNSTABLE_NEGATIVE_TPD	2.79032e-3	—
8	3	Extract	-3.01681e-2	UNSTABLE_NEGATIVE_TPD	4.22071e-3	—
8	4	Raffinate	-1.32498e-2	UNSTABLE_NEGATIVE_TPD	1.91641e-3	—
8	4	Extract	-1.32498e-2	UNSTABLE_NEGATIVE_TPD	2.92731e-3	—
8	5	Raffinate	-1.44561e-17	STABLE_NO_NEGATIVE_TPD	1.27501e-3	—
8	5	Extract	-5.39617e-18	STABLE_NO_NEGATIVE_TPD	1.95491e-3	—
8	6	Raffinate	1.64048e-18	STABLE_NO_NEGATIVE_TPD	8.02741e-4	—
8	6	Extract	1.39131e-16	STABLE_NO_NEGATIVE_TPD	1.23526e-3	—
8	7	Raffinate	-1.80640e-18	STABLE_NO_NEGATIVE_TPD	4.55406e-4	—
8	7	Extract	-1.07661e-18	STABLE_NO_NEGATIVE_TPD	7.03354e-4	—
8	8	Raffinate	8.64656e-18	STABLE_NO_NEGATIVE_TPD	1.97380e-4	—
8	8	Extract	9.19300e-19	STABLE_NO_NEGATIVE_TPD	3.07143e-4	—
9	1	Raffinate	-5.46056e-2	UNSTABLE_NEGATIVE_TPD	5.57756e-3	—
9	1	Extract	-5.46056e-2	UNSTABLE_NEGATIVE_TPD	7.27688e-3	—
9	2	Raffinate	-4.59610e-2	UNSTABLE_NEGATIVE_TPD	4.01653e-3	—
9	2	Extract	-4.59610e-2	UNSTABLE_NEGATIVE_TPD	5.87701e-3	—
9	3	Raffinate	-3.62809e-2	UNSTABLE_NEGATIVE_TPD	2.85177e-3	—
9	3	Extract	-3.62809e-2	UNSTABLE_NEGATIVE_TPD	4.30271e-3	—
9	4	Raffinate	-2.40962e-2	UNSTABLE_NEGATIVE_TPD	1.99350e-3	—
9	4	Extract	-2.40962e-2	UNSTABLE_NEGATIVE_TPD	3.03776e-3	—
9	5	Raffinate	-7.18116e-3	UNSTABLE_NEGATIVE_TPD	1.36430e-3	—
9	5	Extract	-7.18116e-3	UNSTABLE_NEGATIVE_TPD	2.08830e-3	—
9	6	Raffinate	2.14546e-18	STABLE_NO_NEGATIVE_TPD	9.04955e-4	—
9	6	Extract	9.24209e-18	STABLE_NO_NEGATIVE_TPD	1.38837e-3	—
9	7	Raffinate	-1.23199e-17	STABLE_NO_NEGATIVE_TPD	5.68604e-4	—
9	7	Extract	-1.20795e-18	STABLE_NO_NEGATIVE_TPD	8.74651e-4	—
9	8	Raffinate	3.37021e-18	STABLE_NO_NEGATIVE_TPD	3.21360e-4	—
9	8	Extract	-1.10831e-17	STABLE_NO_NEGATIVE_TPD	4.96871e-4	—
9	9	Raffinate	4.31172e-18	STABLE_NO_NEGATIVE_TPD	1.38890e-4	—

N_T	Stage	Phase	TPD minimum	TPD verdict	Local min. eigenvalue	Local verdict
9	9	Extract	1.68109e-19	STABLE_NO_NEGATIVE_TPD	2.16567e-4	—
10	1	Raffinate	-5.57421e-2	UNSTABLE_NEGATIVE_TPD	5.59573e-3	—
10	1	Extract	-5.57421e-2	UNSTABLE_NEGATIVE_TPD	7.26562e-3	—
10	2	Raffinate	-4.83875e-2	UNSTABLE_NEGATIVE_TPD	4.04674e-3	—
10	2	Extract	-4.83875e-2	UNSTABLE_NEGATIVE_TPD	5.90202e-3	—
10	3	Raffinate	-4.05929e-2	UNSTABLE_NEGATIVE_TPD	2.89293e-3	—
10	3	Extract	-4.05929e-2	UNSTABLE_NEGATIVE_TPD	4.35619e-3	—
10	4	Raffinate	-3.13371e-2	UNSTABLE_NEGATIVE_TPD	2.04474e-3	—
10	4	Extract	-3.13371e-2	UNSTABLE_NEGATIVE_TPD	3.11116e-3	—
10	5	Raffinate	-1.93232e-2	UNSTABLE_NEGATIVE_TPD	1.42488e-3	—
10	5	Extract	-1.93232e-2	UNSTABLE_NEGATIVE_TPD	2.17715e-3	—
10	6	Raffinate	-2.39142e-3	UNSTABLE_NEGATIVE_TPD	9.73064e-4	—
10	6	Extract	-2.39142e-3	UNSTABLE_NEGATIVE_TPD	1.49032e-3	—
10	7	Raffinate	8.96347e-18	STABLE_NO_NEGATIVE_TPD	6.43338e-4	—
10	7	Extract	1.37748e-18	STABLE_NO_NEGATIVE_TPD	9.87874e-4	—
10	8	Raffinate	-4.91172e-15	STABLE_NO_NEGATIVE_TPD	4.03848e-4	—
10	8	Extract	-8.93717e-18	STABLE_NO_NEGATIVE_TPD	6.21281e-4	—
10	9	Raffinate	6.66823e-18	STABLE_NO_NEGATIVE_TPD	2.28151e-4	—
10	9	Extract	8.62131e-19	STABLE_NO_NEGATIVE_TPD	3.52332e-4	—
10	10	Raffinate	1.17326e-17	STABLE_NO_NEGATIVE_TPD	9.85580e-5	—
10	10	Extract	-5.45908e-18	STABLE_NO_NEGATIVE_TPD	1.53265e-4	—

Appendix B — product metrics and target summary

N_T	NMP-free recovery %	Saturates %	Total aromatics %	Polar aromatics %	NMP raffinate %	Target outcome
1	91.299	89.980	9.029	0.991	7.271	1 pass / 4 fail / 1 NC
2	88.790	92.023	7.400	0.578	6.981	2 pass / 3 fail / 1 NC
3	87.504	93.101	6.535	0.363	6.835	3 pass / 2 fail / 1 NC
4	86.748	93.746	6.016	0.238	6.749	3 pass / 2 fail / 1 NC
5	86.267	94.160	5.679	0.161	6.695	3 pass / 2 fail / 1 NC
6	85.945	94.439	5.451	0.110	6.658	3 pass / 2 fail / 1 NC
7	85.721	94.633	5.290	0.076	6.633	3 pass / 2 fail / 1 NC
8	85.562	94.772	5.174	0.053	6.615	3 pass / 2 fail / 1 NC
9	85.447	94.873	5.089	0.038	6.601	3 pass / 2 fail / 1 NC
10	85.361	94.947	5.026	0.027	6.592	3 pass / 2 fail / 1 NC

Provenance and limitations

Source identity: Project 223, design 256, job a72e53d4-6747-4bc1-a36f-80bffdf2dc18, completed 2026-09-01T10:30:39.831. Task206 parameter-vector SHA-256: 4b778e4a3695abec88742b8b4987e1fa11695baede98049dadb346014c556023.

- Task206 frozen validation is INCOMPATIBLE_FAIL_CLOSED.
- No active-model global qualification is attributed to Task206.
- Formal COSMO-SAC sulfur remains NOT_CALCULABLE.

The replay uses the existing recorded results only. Worker-carried active-model global qualification was segregated and is not attributed to Task206. Numerical closure, branch reproduction, and absence of a replay-stage TPD blocker do not confer acceptance.